A Decade in a Day: Synthesizing Agent-Native Interfaces from Human-Oriented Software Evolution

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Paper version: arXiv 2610.15765v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *What constitutes a good interface for agents interacting with desktop applications?*

GUIs and SDKs were built for **humans and developers**. This work uses **programmatic** interfaces, not screens.

Motivation: three interface properties

1. **Discoverability.** Discovery takes 24.1–36.2% of tokens.
2. **Granularity.** Long chains of fragile low-level calls.
3. **Observability.** Writes without matching reads.

Evaluation setup

DeskIF-Bench · 255 → 368 tasks | 8 desktop apps | DeepSeek V4 Flash | Claude Haiku 4.5 | OpenCode · pass@3

- **Online:** script verifiers on saved artifacts; **no screenshots.**
- **Held out:** Calc and GIMP unseen in development.

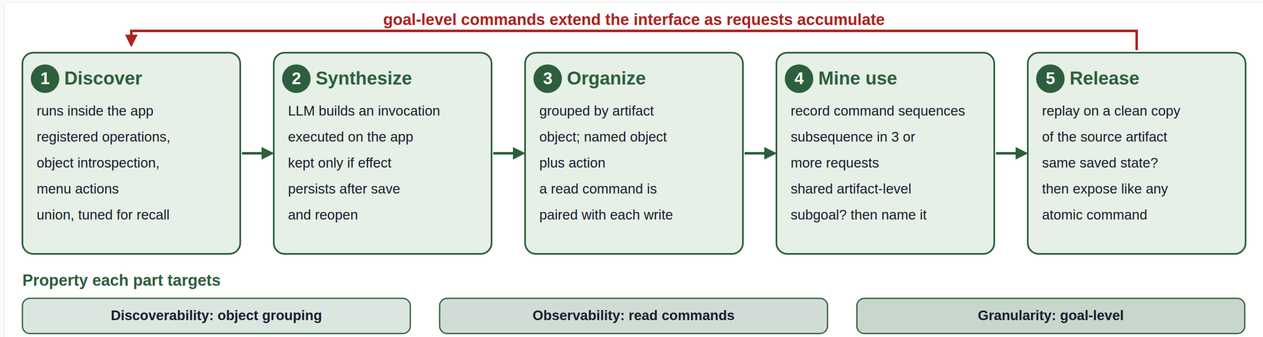
Our read (CUA Radar)

66	★★	online
score of 100	band	evidence level

- **Reports:** artifact-verified online eval; two backbones; baselines; observation ablation; code.
- **Flags:** AP5, goal-level commands built from tasks later tested. **Not reported:** variance.
- Pattern K29. Authors' numbers, not rerun.

Method: ANIS synthesizes atomic commands, then goal-level ones from use

- **Sees / acts:** text and files; shell-style commands on the running app.
- **Atomic:** found in the live app, kept only if the effect survives save and reopen.
- **Goal-level:** sequences recurring in 3 or more requests, released after a clean replay.
- **Build cost:** 17.1–36.4 hours per application, separate from online tokens.



Our redraw (not the paper's figure): the two-phase synthesis of Section 5.

Key results: overall success (%), DeepSeek V4 Flash

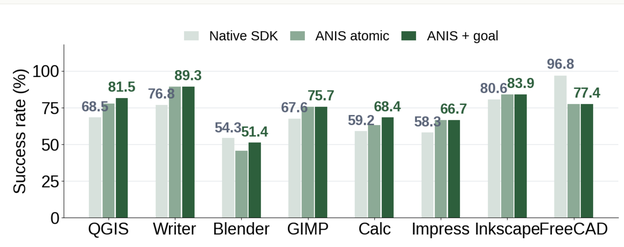
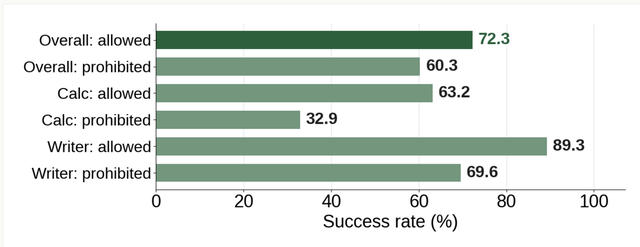


Table 2, redrawn: 368 tasks, pass@3; no variance reported. CLI-Anything (17.4% overall) is not charted.

Overall: ANIS 72.3 / 74.5 vs native SDK 68.5

SO WHAT → *Tokens per task fall 73.2% / 74.9%. Native SDK is higher on FreeCAD (96.8 vs 77.4) and Blender (54.3 vs 45.7). 98.7% of CLI-Anything's failures are missing commands. Haiku 4.5 keeps the order: native 61.4, ANIS 64.9 and 67.4.*

Observation commands, DeepSeek V4 Flash (%)



Appendix D, Table 5, redrawn: 368 tasks, atomic interface; observation prohibited by instruction.

SO WHAT → *Prohibiting observation commands lowers overall success by 12.0 points. Calc loses 30.3 and Writer 19.7, while GIMP, Impress and Inkscape are unchanged. An audit of the saved trajectories found no observation command used. In a Blender task, an agent that could not read the chair's orientation applied a 180° rotation that was accepted but failed verification.*

Headline Numbers

74.5%
success, 8 applications
vs 68.5% native SDK, DeepSeek

- **73.2%** fewer tokens, atomic
- **17.4%** CLI-Anything success
- **67.4%** Haiku with goal-level

The paper's own controls

CONTROL	ANIS	COMPARED WITH
Haiku 4.5, overall	64.9 / 67.4	native SDK 61.4
Held-out tasks (110)	60.9% (67/110)	atomic 58.2% (64/110)
Held-out apps Calc / GIMP	68.4 / 75.7	native 59.2 / 67.6
Docs on native SDK	-1.9 to +1.4 pts	no consistent gain

Takeaway

Synthesized commands raise success from 68.5% to 74.5% and cut tokens by about 73%, with FreeCAD and Blender below the native SDK; build cost and variance are not reported.

17.1–36.4 h to generate | 10 goal-level commands | 12 of 12 series: new > old

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Paper | Code